KuoWei Tseng

Electrical Engineer | AI & Machine Learning | Embedded Systems & Robotics |
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EDUCATION

UNIVERSITY Of CALGARY **Calgary, CA**
Bachelor of Science in Electrical Engineering (Digital Engineering Minor) CGPA: 3.3/4.0

Relevant coursework: Deep Learning, Embedded Systems, Digital Signal Processing, VLSI Design, Control Systems

CERTIFICATE & PROFESSIONAL DEVELOPMENT

PyTorch for Deep Learning Specialization (Mar, 2026) DeepLearning.AI

Natural Language Processing with Sequence Model (Mar, 2026) DeepLearning.AI

Natural Language Processing with Attention Model (May, 2026) DeepLearning.AI

PLC Technician Certificate I (Expected completion: May 2026). George Brown College, Toronto, ON

- Ladder Logic programming fundamentals for industrial control system.
- PLC hardware: CPU, I/O interfaces, timers, counters, state-based control
- HMI integration and troubleshooting practices

SKILLS

Core:
Python, PyTorch, TensorFlow, fastai, Matplotlib, OpenCV, Git, Linux, NumPy, Pandas, C

Familiar:
Java, SystemVerilog, VHDL, LTspice, Cadence Virtuoso

Embedded & Control:
Raspberry Pi 5, Jetson Orin Nano, Pi Pico, ESP32, PIC Microcontroller, PLC (Ladder Logic)

WORK EXPERIENCE

Lision Technology Inc.
Taoyuan, Taiwan
R&D Assistant intern
06/2024 – 08/2024

- Characterized 15+ discrete semiconductor devices (PN diodes, Zener, Schottky, TVS, MOSFETs) by measuring I-V curves, breakdown voltage, leakage current, and recovery time against datasheet specifications.
- Built a Python data pipeline to collect, process, and visualize electrical measurement data across device test cases, reducing manual tabulation time.
- Evaluated thermal and mechanical packaging trade-offs (flat-lead vs. bent-lead, solder vs. wire bonding) to support power-density optimization decisions.
- Assisted in IC performance verification against design specifications, identifying measurement inconsistencies during characterization.

- Gained exposure to semiconductor packaging manufacturing flow including DO, SOD, SOT, and TO package standards.

ACADEMIC PROJECTS

Project Name: ZEROG COATING SYSTEM—Capstone Project (https://www.zerogcoating.com)

- Developed a corrosion detection model (ResNet-18 + cosine similarity prototypes) achieving 95.6% test accuracy and 0.873 OOD AUROC on real-world industrial data
- Designed a cosine similarity–based rejection mechanism to flag out-of-distribution inputs, improving reliability under domain shift
- Built a real-time perception-control loop fusing LiDAR and RGB inputs for autonomous detect → classify → react decisions on Raspberry Pi 5
- Architected a multi-MCU system (Raspberry Pi 5 + Pi Pico + ESP32) separating high-level AI inference from low-latency actuation
- Engineered LiDAR-based obstacle avoidance achieving 94% collision avoidance rate; encoder-based motion control held drift to ~1.46°/meter
- Delivered a fully integrated prototype with ~2.5hr operational runtime under normal conditions

Project Name: Representation Learning & OOD Robustness Study
(https://github.com/Tseng0318/ENDG511project.git)

- Benchmarked four representation learning methods (SimCLR, SupCon, BYOL, ViT) on a shared ResNet-18 backbone, evaluating each on in-distribution accuracy, OOD detection AUROC, and domain shift robustness (grayscale)
- Implemented cosine similarity–based prototype OOD detection, achieving 0.839 AUROC and ~68% rejection rate with SimCLR — outperforming supervised baselines on both metrics
- Showed SimCLR's feature space generalizes better under domain shift than supervised contrastive and cross-entropy training, with measurable accuracy retention on grayscale test splits
- Designed and evaluated an early-exit inference framework
- Applied low-rank factorization to compress the ResNet-18 backbone, quantifying the accuracy-vs-parameter-count tradeoff for edge deployment feasibility

Project Name: TrashAuto—Autonomous Garbage Collection Rover
(https://github.com/Tseng0318/yolo_waste.git)

- Trained and developed a two-stage CNN pipeline (YOLO and ResNet34) to classify garbage versus non-garbage objects. (Reached 91.7% real world accuracy)
- Designed and built an autonomous rover integrating LiDAR, ultrasonic sensors, IMU, wheel encoders, and camera
- Implemented sensor-based object detection and distance tracking for navigation and obstacle avoidance using LiDAR
- Optimized real-time inference on Raspberry Pi by event-driven triggering to reduce computation load

PUBLICATION

1. Feasibility of recent peptide therapy for ischemic stroke. K.-F. Tseng, **K.-W. Tseng**, et al. Journal of Pharmacological Sciences, 2026. (Contribution: Developed Python-based data analysis pipeline for computational methodology)